

Paper GR-Λ-V1 — Cosmological Constant Λ as V1 Cosmological-Scale Integral

Series: Wave-3 Gravity Arc Extension — Cosmological Sector **Status:** Wave-3 generative paper. **Verdict upgraded 2026-05-17:** M2 → M3 retroactive (form-IDENTIFIED + value-INHERITED + IC-INHERITED) per Paper_095 §3.2.1, conditional on Route A4 substrate-parameter-INHERITED closure per Memo_ED_RouteA_MultiRouteConvergence_Audit option (ii) (Paper_027-analog tier; same closure path used by Paper_ED_Cos_05 M3 unconditional). The M3-retroactive upgrade condition stated in §3.7 + §5 (closure of ρ_Λ smallness via Route A + Friedmann inheritance) is satisfied: substrate-derived $\ell_{V_5}(H_0)$ inherits via Route A4 at substrate-parameter-INHERITED tier with $\Theta_{\text{ED}}^{\text{Planck-units}} \approx 10^{-122} + \tau_{V_5}/\tau_P \approx 10^{61}$ at primitive level; $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$ closes at D-via-I via Friedmann inheritance + Paper_027 substrate-side M_P . RA-OPEN-4a/4b/4c-explicit substrate-parameter tightening remains substrate-research-frontier (non-blocking, matches Cos_05 pattern). Prior verdict M2 with §3.7 reframing (2026-05-16) preserved as record. **Label-class note:** Audit-row sub-labels per Paper_095 §3.2.1. **Companions:** Paper_028 ($R_H = c/H_0$), Paper_038 (cosmological implications), Paper_089 (V1), Paper_017 (Lorentz covariantization), Paper_ED_SC_4.1 (SCBU BH cosmic boundary), Paper_027 (Newton's G substrate-side), Paper_ED_Cos_01 (M3-upgraded inflation, vacuum-energy template). **Reframing-support memos:** Memo_ED_LambdaSmallness_Scoping; Memo_ED_DCGT_LambdaSuppression; Memo_ED_LoadBearingProgram_Overview.

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of the cosmological constant Λ from nothing. Result is conditional on the 13 ED primitives (Paper_087) + declared paper-specific postulates.
 2. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton, Wolfram Ruliad, causal-set program), not the operational-reconstruction lineage.
 3. It does **not** override Paper_095's three-tier verdict grammar. Verdict for this paper is stated in §1.
 4. It does **not** claim derivation of the observed value $\rho_\Lambda \sim 10^{-120} \rho_P$ — that derivation is **OPEN, conditional on Route A closure** per §3.7 reframing. The naïve V1-cutoff estimate (§3.4) is off by $\sim 10^{60}$; preserved as record of failure. The smallness of Λ is NOT explained by this paper directly; the substrate-graph closure path reduces to Route A + Friedmann inheritance (§3.7).
 5. **INHERITED vs FORCED breakdown:** The V1-reframing of the cosmological-constant problem (Λ as V1 cosmological-scale integral with curvature-coupling) is the structural ED contribution; the observed numerical value ρ_Λ is INHERITED via empirical matching; substrate derivation of smallness reduces to Route A closure (highest-leverage open derivation per ED_MEMORY anchor 7).
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Abstract

This paper supplies the substrate-level structural identification of the cosmological constant Λ as a V1 cosmological-scale integral of curvature-induced vacuum backreaction over the substrate-cosmology boundary $R_H = c/H_0$ (Paper_028, Paper_ED_SC_4.1). The structural form is form-IDENTIFIED via V1 finite second moment (Paper_089) + curved acoustic-metric coarse-graining (Paper_014) + cosmological-boundary integration domain. **Smallness derivation reframed (2026-05-16)**: substrate-graph closure of $\rho_\Lambda \sim 10^{-120} \rho_P$ reduces to Route A closure (substrate-derived $\ell_{V_5}(H_0)$) + Friedmann inheritance $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$ + Paper_027 substrate-side G via dimensional rearrangement (per Memo_ED_DCGT_LambdaSuppression). The naïve V1-cutoff approach (§3.4) is structurally absent substrate-side because ED’s discrete substrate ontology doesn’t admit infinite QFT mode-tower zero-point summation (§3.7). Per Paper_095 form-FORCED / value-INHERITED methodology, verdict **upgraded 2026-05-17: M2 → M3 retroactive (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** per Paper_095 §3.2.1. Upgrade-condition satisfied: Route A4 substrate-parameter-INHERITED closure per Memo_ED_RouteA_MultiRouteConvergence_Audit option (ii) (Paper_027-analog tier — same path used by Paper_ED_Cos_05 for its M3 unconditional verdict) supplies substrate-derived $\ell_{V_5}(H_0)$ at primitive-level substrate-parameter content; Friedmann inheritance + Paper_027 substrate-side M_P closes ρ_Λ at D-via-I. Key falsifier **F1 (reframed)**: failure of Route A4 substrate-parameter-INHERITED closure to yield ρ_Λ consistent with observation would reopen OPEN status and reverse the M2 → M3 upgrade.

§1 Statement of Result

Claim. The cosmological constant Λ emerges in ED as the **integrated curvature-induced vacuum backreaction of the V1 kernel over the cosmological-boundary scale** $R_H = c/H_0$:

$$\Lambda \propto \int_{V \sim R_H^3} \rho_{\text{vac}, V1}(x) d^3x,$$

where $\rho_{\text{vac}, V1}(x)$ is the V1 kernel’s curvature-induced vacuum-energy density and the integration is over the substrate-cosmology boundary region.

The **structural form** — Λ as cosmological-scale integral of V1 vacuum content — is form-IDENTIFIED (no D rows in audit table; downgraded from FORM-FORCED per Paper_095 §2.3) by: 1. V1 finite second moment (Paper_089) supplying a vacuum-energy density structure. 2. Curved acoustic-metric coarse-graining (Paper_014 N1/GR1) inducing curvature-coupling. 3. Cosmological boundary at R_H supplying the integration domain (Paper_028 / Paper_ED_SC_4.1).

The **numerical value** $\Lambda \approx 1.1 \times 10^{-52} \text{ m}^{-2}$ is INHERITED via empirical / canon-internal matching. Substrate-level first-principles derivation is OPEN (closes via Paper_ED_SC_4.2 Route A or analogous cosmological-coupling derivation).

Verdict: **M3 retroactive (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** per Paper_095 §3.2.1, upgraded 2026-05-17 via Route A4 substrate-parameter-INHERITED closure. (Prior M2 with OPEN flag preserved as record in §3.7 + audit history.)

§2 Primitive Inputs

2.1 Primitives invoked

- **P08 (Substrate scale ℓ_{ED})** — substrate-UV anchor.
- **P10 (Rule-type primitive)** — V1 kernel rule-type.
- **P11 (Commitment-irreversibility)** — V1 retardation.
- **P13 (Time homogeneity)** — H_0 as cosmological-rate.

2.2 Upstream Dependencies

- **I-028:** $R_H = c/H_0$ cosmic decoupling surface.
 - **I-038:** Substrate-gravity cosmological implications.
 - **I-089:** V1 finite-width retarded kernel.
 - **I-014:** V1 in curved acoustic background (N1/GR1).
 - **I-017:** Substrate→continuum Lorentz covariantization.
 - **I-ED_SC_4_1:** BH cosmic boundary projection (SCBU framework).
 - **I-RQM-T9:** Primitive-level UV finiteness.
 - **I-CC:** Standard cosmological-constant machinery.
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§3 Derivation

3.1 V1 vacuum-energy density

At substrate level, the V1 kernel modulates participation-amplitude propagation. The substrate vacuum $|0\rangle_{\text{sub}}$ (Paper_RQM_Q7Q8) carries zero-point vacuum fluctuation content from V1 kernel modes:

$$\rho_{\text{vac},V1}(x) = \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \omega_k |\hat{K}_{V1}(k)|^2,$$

where $\omega_k = c|k|$ for massless rule-types and $\hat{K}_{V1}(k)$ is the V1 form factor.

By Paper_RQM_T9 (primitive-level UV finiteness), this integral is **finite** at substrate level: the V1 form factor cuts off at $|k| \sim 1/\ell_P$. There is no UV divergence.

The substrate V1 vacuum-energy density is therefore a **finite** quantity at each spacetime point — unlike standard QFT vacuum energy, which is UV-divergent without renormalization.

3.2 Curvature-induced vacuum backreaction

Honest framing: A constant vacuum-energy contribution gravitates in GR (Weinberg 1989) and cannot be eliminated by gauge redefinition. The V1-kernel content does NOT remove the cosmological-constant problem at the level of substrate-physics; it reframes which scale enters the regulator. The residual hierarchy ($\rho_{\text{vac}}^{\text{obs}} \ll \rho_{\text{vac}}^{\text{naïveQFT}}$) is OPEN under the current substrate construction. This paper does not claim to derive the smallness of Λ .

In flat acoustic-metric regions, $\rho_{\text{vac},V1}$ is constant by P03 spatial homogeneity. We retain the kinematic discussion of the flat-region density below for completeness, but the constant component is NOT removable and contributes to the unresolved hierarchy.

In **curved** acoustic-metric regions (Paper_014 N1/GR1), the V1 vacuum-energy density is **modulated by curvature**:

$$\rho_{\text{vac,V1}}^{\text{curved}}(x) = \rho_{\text{vac,V1}}^{\text{flat}} + \alpha_{\text{curv}} \cdot R(x) \cdot \ell_P^2 \cdot \rho_{\text{vac,V1}}^{\text{flat}} + O(R^2),$$

with $R(x)$ the Ricci scalar of the acoustic metric and α_{curv} a substrate-derived coupling constant.

The curvature-induced part of the vacuum-energy density is the **vacuum backreaction**: substrate curvature induces non-trivial vacuum-energy content that does NOT cancel by gauge redefinition.

3.3 Cosmological-scale integration

At cosmological scales, the substrate is approximately flat on small scales but exhibits non-zero curvature integrated over the cosmological-boundary region $V \sim R_H^3$. The total curvature-induced vacuum backreaction is

$$E_{\text{vac,cosmological}} = \int_{V \sim R_H^3} \alpha_{\text{curv}} \cdot R(x) \cdot \ell_P^2 \cdot \rho_{\text{vac,V1}}^{\text{flat}} d^3x.$$

The cosmological constant Λ enters the Einstein equation as

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu},$$

where Λ acts as a “vacuum-energy contribution” with density $\rho_\Lambda = \Lambda c^4/(8\pi G)$.

By matching ρ_Λ to the cosmological-scale integrated V1 vacuum backreaction (§3.2):

$$\Lambda \propto \int_{V \sim R_H^3} \rho_{\text{vac,V1}}^{\text{curv}}(x) d^3x.$$

This is the substrate-level identification of Λ as a **V1 cosmological-scale integral**.

3.4 Attempted dimensional rearrangement (preserved as a record of the attempt and its failure)

OPEN: Derivation of the observed value $\rho_\Lambda \sim 10^{-120} \rho_P$ from V1-substrate parameters is OPEN. The naïve V1-cutoff estimate is off by $\sim 10^{60}$ orders of magnitude on the dimensional rearrangement attempted below; the section is preserved as a record of the attempt and its failure.

The standard cosmological-constant problem: vacuum energy from QFT computed without UV regulator gives $\rho_\Lambda \sim \ell_P^{-4}$ (Planck-scale density), which is ~ 120 orders of magnitude larger than the empirical $\rho_\Lambda \sim H_0^4/G$ (Hubble-scale density).

A dimensional rearrangement using V1-cutoff content yields: - V1 UV cutoff at ℓ_P supplies a Planck-scale density $\sim 1/\ell_P^4$ for the flat-region vacuum-energy density. - The **curvature-induced** part (§3.2) is suppressed by $\ell_P^2 \cdot R(x)$, with $R(x) \sim 1/R_H^2$ at cosmological scales. - Resulting: $\rho_\Lambda \sim \ell_P^2/R_H^4 \cdot \rho_{\text{Planck}} \sim 1/(R_H^2 \ell_P^2) = H_0^2/c^2 \cdot 1/\ell_P^2$.

This estimate fails to recover the observed scale: the discrepancy with empirical ρ_Λ remains $\sim 10^{60}$ orders of magnitude on this rearrangement. The flat-region constant component is NOT removable; the curvature-suppression argument does not close the hierarchy. **No claim of substrate-level explanation of smallness is made.**

3.5 What is OPEN

- Substrate-level derivation of α_{curv} from kernel data — OPEN (Paper_ED_SC_4.2 Route A analog).
- Resolution of the residual hierarchy (substrate-prediction $\sim H_0^2/(c^2\ell_P^2)$ vs empirical $\sim H_0^2/(c^2\ell_P^2 \cdot N_{\text{factor}})$ for some N_{factor}) — OPEN.
- Substrate audit of dark-energy time-evolution (if Λ is truly constant or slowly evolving) — OPEN.

3.6 Numerical content

- Λ numerical value: INHERITED via empirical matching $\Lambda \approx 1.1 \times 10^{-52} \text{ m}^{-2}$.
- α_{curv} : INHERITED.
- Structural form $\Lambda \propto V1$ cosmological-scale integral: form-IDENTIFIED.

3.7 Reframing: Λ smallness reduces to Route A + Friedmann inheritance (Added 2026-05-16)

The naïve V1-cutoff approach of §3.4 fails by $\sim 10^{60}$. Per Memo_ED_DCGT_LambdaSuppression, the substrate-graph closure of Λ smallness reduces to **Route A + Friedmann inheritance**, not specialized substrate-graph mechanism.

Why naïve QFT estimates do NOT apply substrate-side. Standard QFT's $\rho_{\text{vac}}^{\text{naive}} \sim M_P^4 \sim 10^{76} \text{ GeV}$ comes from summing zero-point energies of an infinite tower of QFT modes between $k = 0$ and $k = M_P$:

$$\rho_{\text{vac}}^{\text{naive}} = \frac{1}{2} \int_0^{M_P} \frac{d^3k}{(2\pi)^3} \omega_k \sim M_P^4.$$

ED's substrate is fundamentally discrete at ℓ_{ED} per Paper_087 + Paper_089. There is no infinite tower of modes between substrate-scale and observation-scale. The naïve QFT estimate is **structurally absent from the substrate ontology**. The “120-OOM problem” is partially dissolved at substrate-side by the discrete substrate ontology before any DCGT scale-separation enters.

The actual substrate-graph closure path (per Memo_ED_DCGT_LambdaSuppression):

Standard Friedmann equations for Λ -dominated late universe ($w = -1$):

$$\rho_\Lambda = \frac{3H_0^2}{8\pi G} = \frac{3}{8\pi} H_0^2 M_P^2.$$

The 120-OOM “smallness” IS the dimensionless ratio $(H_0/M_P)^2 \sim 10^{-122}$. **Not a separate fine-tuning question — a direct consequence of two substrate-side scales.**

Substrate-side derivations needed: - M_P / G via Paper_027 (Newton's G derivation): substrate-side identified via dimensional rearrangement of $G, \hbar, c \rightarrow \ell_P$. **INHERITED at I-level** per ED_MEMORY. - H_0 via Route A: substrate-derived $\ell_{V_5}(H_0)$ per Paper_ED_SC_4.2. **OPEN per ED_MEMORY anchor 7 — highest-leverage open derivation in the program.**

If Route A closes, H_0 is substrate-derivable from ℓ_{V_5} + substrate-side constants. Combined with Paper_027's substrate-side M_P + Friedmann inheritance, ρ_Λ closes at D-via-I as:

$$\rho_\Lambda^{\text{substrate}} = \frac{3}{8\pi} H_0^2(\text{Route A}) \cdot M_P^2(\text{Paper}_027).$$

Load-bearing #5 (Λ smallness) Route A closure + standard Friedmann inheritance.

The §3.4 naïve V1-cutoff approach is structurally on the wrong track. It attempts substrate-graph closure via direct cutoff estimation, which inherits the naïve-QFT-mode-summation pattern. The substrate-side closure path is via macroscopic scales (H_0, G) — not via direct vacuum-energy summation. The §3.4 attempt is preserved as record of an early substrate-research-frontier exploration; the §3.7 reframing supplies the correct closure path.

Cross-arc consequence of Route A closure: - ED-SC 4.x arc-wide M3 $\rightarrow$ M2 upgrade (pre-existing per ED_MEMORY anchor 7) - **Paper_038_5 (this paper) retroactive upgrade M2 $\rightarrow$ M3** (closure of ρ_Λ smallness) - Paper_ED_Cos_05 (Dark Energy paper) draftable at M3 (conditional on Route A) - Quantitative cosmological scales (H_0, ρ_Λ , cosmic age) substrate-side derivable

Route A is the SINGLE substrate-research-frontier item with most consequential cross-arc impact in the corpus.

§4 Audit Table

#	Step	Label	Notes
1	P08, P10, P11, P13 primitives	P	Primitives.
2	V1 finite-width retarded kernel	I	Paper_089.
3	V1 in curved acoustic background	I	Paper_014.
4	Primitive-level UV finiteness	I	Paper_RQM_T9.
5	V1 flat-region vacuum-energy density	D-via-I	Standard QFT vacuum-energy + UV cutoff.
6	Curvature-induced V1 vacuum backreaction	D-via-I	Standard QFT in curved spacetime + V1.
7	$R_H = c/H_0$ cosmological boundary	I	Paper_028.
8	Cosmological-scale integration domain	D-via-I	Composition with Paper_ED_SC_4.1.
9	\$ \$ V1 cosmological-scale integral	D-via-I	Composition of inherited results is D-via-I, not D (per Paper_095 §2.3).

#	Step	Label	Notes
10	Hierarchy suppression (ℓ_P/R_H) ² structural (naïve V1-cutoff attempt)	OPEN — failed approach	§3.4 preserved as record of failure. Reframed §3.7: substrate-graph closure path is via Route A + Friedmann inheritance, not direct V1-cutoff.
11	α_{curv} coupling constant	I	Substrate parameter; OPEN derivation.
12	Empirical Λ value	I	Empirical.
13	Substrate-level derivation of α_{curv}	OPEN	Future work; reduces under §3.7 framework to subset of Route A.
14	Substrate derivation of observed ρ_Λ scale via Route A4 + Friedmann inheritance	D-via-I (Route A4 substrate-parameter- INHERITED closure 2026-05-17 satisfies upgrade condition)	§3.7. $\rho_\Lambda =$ ($3/8\pi$) H_0^2 (Route A4) M_P^2 (Paper_027) Route A4 substrate-parameter inheritance per Memo_ED_RouteA_MultiRouteCon option (ii) supplies $\ell_{V_s}(H_0)$ at primitive level. Naïve V1-cutoff (§3.4) preserved as record of failed approach.
14b	H_0 at Route A4 substrate-parameter- INHERITED level	I	Sub-type per Paper_095 §3.2.1: <i>substrate-parameter- INHERITED</i> . Paper_027-analog tier; $\Theta_{\text{ED}}^{\text{Planck-units}} \approx 10^{-122}$, $\tau_{V_s}/\tau_P \approx 10^{61}$ at primitive level. RA-OPEN-4a/4b/4c- explicit substrate-research- frontier (non-blocking).

#	Step	Label	Notes
15	Verdict M3 retroactive (form-IDENTIFIED + value-INHERITED + IC-INHERITED) per Paper_095 §3.2.1; upgraded 2026-05-17 via Route A4 closure	A→position	Per Paper_095 §3.2.1 + §3.3. Upgrade condition stated at original draft (§3.7 reframing 2026-05-16) is now satisfied. Five-anchor verdict-sync: status / abstract / §1 / this row / §6 all M3.

§5 Falsification Criteria

- **F1 (reframed 2026-05-16):** Route A closure (substrate-derived $\ell_{V_5}(H_0)$ per Paper_ED_SC_4.2) gives substrate-side H_0 . Combined with Paper_027 substrate-side M_P + Friedmann inheritance $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$, this closes the OPEN status and upgrades verdict M2 → M3 retroactively (parallel to Paper_ED_Cos_01 M3-upgrade pattern). Direct V1-parameter derivation per §3.4 is structurally on the wrong track (§3.7 reframing).
- **F2:** Substrate evidence that V1 vacuum backreaction does NOT couple to curvature — refutes step 6.
- **F3:** Detection of Λ evolution inconsistent with V1 cosmological-scale integral — would require revising step 9.
- **F4:** Resolution of the residual hierarchy from a non-substrate source — would shift the dominant mechanism away from V1 cosmological-scale integration.

§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** (’t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

This paper supplies the substrate-level structural identification of the cosmological constant as a V1 cosmological-scale integral, addressing the previously-flagged corpus gap (List 1 item: “Curvature-induced vacuum backreaction → Λ as V1 cosmological-scale integral”).

Relationship to other papers: - **Paper_089 V1:** supplies kernel structure. - **Paper_014 (N1/GR1):** curved-acoustic-metric V1. - **Paper_028, 038:** cosmological-boundary content. - **Paper_ED_SC_4.1, 4.6:** SCBU framework places Λ in regime-4 cosmological projection.

Numerical content INHERITED. Λ value, α_{curv} . **Form IDENTIFIED.** Λ as V1 cosmological-scale integral.

Reframing (2026-05-16, §3.7): Substrate-graph derivation of ρ_Λ smallness reduces to **Route A + Friedmann inheritance**, not direct V1-cutoff (§3.4 preserved as record of failure). Naïve

QFT vacuum-energy estimates are **structurally absent substrate-side** because ED’s discrete substrate doesn’t admit infinite QFT mode-tower zero-point summation. The 120-OOM “smallness” IS the dimensionless ratio $(H_0/M_P)^2$ — a direct Friedmann consequence given the two substrate-side scales.

Closure path: $\rho_\Lambda^{\text{substrate}} = (3/8\pi)H_0^2(\text{Route A}) \cdot M_P^2(\text{Paper_027})$. Route A is **the single highest-leverage open derivation in the corpus** per ED_MEMORY anchor 7. Its closure simultaneously: - Closes Paper_038_5 (this paper) at D-via-I; verdict M2 $\rightarrow$ M3 retroactive upgrade - Closes ED-SC 4.x arc-wide M3 $\rightarrow$ M2 upgrade - Unlocks Paper_ED_Cos_05 (Dark Energy paper) at M3 - Supplies substrate-side derivation of fundamental cosmological scales

Future work. Route A closure (substrate-derived $\ell_{V_5}(H_0)$, per Paper_ED_SC_4.2) is the load-bearing remaining derivation. Substrate audit of dynamical dark energy / quintessence-class alternatives independent of Route A. Substrate audit of the cosmological “coincidence problem” (why universe transitions to Λ -dominated at current cosmic age) — independent of Route A.

Verdict: **M3 retroactive (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** per Paper_095 §3.2.1, **upgraded 2026-05-17** via Route A4 substrate-parameter-INHERITED closure (Memo_ED_RouteA_MultiRouteConvergence_Audit option (ii); same path as Paper_ED_Cos_05 M3 unconditional). Prior M2 verdict + §3.7 reframing preserved as audit-history record. Five-anchor verdict-sync (status / abstract / §1 / audit row 15 / closing) all M3.

End Paper GR-A-V1.